

EXAMINERS REPORT ON Ph.D. THESIS

Title of thesis Application of Graph Theory in TransportationName of candidate Shri/Smt./Ku. Sanjay Kumar BisenSubject: mathematicsFaculty Science# 1. Thesis is recommended for the Yes/~~No~~ Award of Ph.D degree..... ✓

2. The thesis be revised on the Lines —

Detailed below

3. The thesis be rejected — (Please write Yes/No, as the case may be)

Thesis requiring only minor revision/s should also be covered in the category and suitable remarks detailing minor revision/s required, is/are to be enumerated. X

Thesis requiring only major revision/s should be covered in this category and suitable remarks detailing the major revision/s required is/are to be enumerated X

DETAILED REPORT

(The examiner is required to attach detailed report in four copies, covering also the following points)

(a) It must be a piece of research work characterized either by the discovery of the facts or by a fresh approach towards the interpretation of facts. In either case it should evince the candidate's capacity for critical examination and sound judgment. Report enclosed.

(b) It must be satisfactory in point of language and presentation of the subject matter.

Date: 22.06.2020Place: Bhopal(Signature and Seal of the Examiner)
Dr. Anil Goyal
22.06.2020Full Name & Address: Prof. Anil GoyalDept. of MathematicsUIT-RGV, Bhopal-33mobile: 9425338282

Department of Mathematics
RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA,
Airport Bypass Road, Gandhi Nagar, Bhopal (M.P.)

Dr. Anil Goyal

Professor, Department of Mathematics

Ph.D. Thesis Report

Report on the thesis entitled "Application of Graph Theory in Transportation" submitted by **Sanjay Kumar Bisen** to Ram Krishna Dharmarth Foundation University, Bhopal (M P) for the award of Ph.D.degree in MATHEMATICS.

The candidate has worked on various types of Transportation problems and methods for solving them with the help of Graph Theory. Also discuss various applications of Transportation problems. The thesis is divided into five chapters.

Chapter-I describes the historical background of various types of graphs, basics about domination in graph theory and applications of graph theory. Also discusses different types of transportation and networking problems.

Chapter -II deals with applications of Graph Theory on different types of transportation problems using graphical representation.

Chapter-III deals with applications of Graph Theory on air transportation problems with results.

Chapter-IV describes applications of graph theory in cyber security data transportation and construction of reliable communication networks.

Chapter-V describes applications of graph theory in railway transportation for railway tracks design , adapting block system.

The thesis is well written and a lot of literature has been collected. The literature survey and literary presentation appears to be up to date. The candidate has published a good number of research papers in the journals of National/International repute.

Based on her work presented in this thesis, I recommend this thesis for the award of the Ph.D. degree in Mathematics of Ram Krishna Dharmarth Foundation University, Bhopal (M P),to **Sanjay Kumar Bisen** after satisfactory performance at the viva-voce examination.

22.06.2020
(Prof. Anil Goyal)
Dr. Anil Goyal
R.G.P.V., Bhopal (M.P.)

EXAMINERS REPORT ON Ph.D. THESIS

Title of thesis Application of Graph theory in Transportation

Name of candidate Shri/Smt./Ku. Sanjay Kumar Bisen

Subject: Mathematics

Faculty Mathematics

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Date: 4/7/2020

Place: Thamir

Badal

(Signature and Seal of the Examiner)

Full Name & Address Dr. DHARMENDRA BADAL

Dept. of Math. & Computer Applications

Bundelkhand University, Thamir

Detail Report

of Thesis of Shri Sanjay Kumar Bisen.

Report is positive and having following points.

- ① The work entitled "Application of Graph Theory in Transportation" As per page no. 222 report Usbund Analysis report the thesis is upto standard both in respect of contents and language for new Mathematics idea.
- ② The chapter 1 is based on road Transport Highway design, road cross section based on road network, graph theory in Highway design.
- ③ The chapter 2 is based on actually Part Two of road Transport representation using graphical terms of road Transportation, graph presentation of line structure and line intersection and representation of segmentation and line structure include road matrix representation.

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- ④ The chapter 3 is based on Application in airline transportation include airline pass problem and air traffic control using graph theory and air port by graph.
- ⑤ The chapter 4 is based on data transportation defining routing using path problem min-cost spanning tree. Travelling salesmen problem and shortest route problem.
- ⑥ The chapter ⑤ includes railway transportation Concept from graph theory in railway network, time table, data and railway network. It also include output data railway network and train track using graph theory and signal network no cycle condition and flow
- ⑦ Reference are not in proper way as per International or national settings.
- ⑧ The thesis must be a piece of research work by a fresh approach towards interpretation of facts.

So there is recommended for award of Ph. D. Degree.

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